ISE 305

Introduction to Database Technology

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Instructors (in alphabetical order):

Ahmad Esmaili (esmaili@cs.stonybrook.edu)

Fusheng Wang (fusheng.wang@stonybrook.edu)

Cheng Zhang (cheng.zhang@ahu.edu.cn)

Lectures

In person classes

- You must be present during lecture
- Silence your cell phone
- Feel free to ask questions

General Class Issues

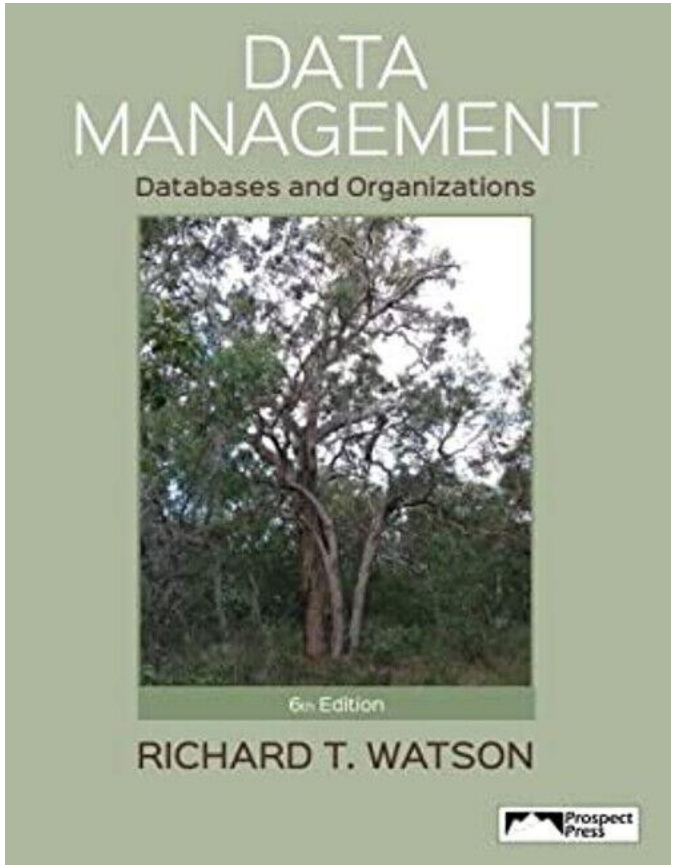
- Hands-on class
- In-class examples
- In-class “Are We on Track” exercises
- MS-Access available on computers in labs
- Software generally available in MS Office

Working exercises during class sessions is critical to your success

Textbook

- Data Management Database and Organizations by Richard T. Watson (open edition)
- https://www.richardtwatson.com/open/

Supplemented with other WEB assigned readings



Purpose

- Understand the use of databases in information systems
- Gain hands-on experience with small-medium sized DBs using a GUI front end
- Learn issues involved in enterprise style systems
- Emphasis on translating requirements into a DB
- Limited emphasis on DB performance

Tailored to DB-related jobs in information systems

Course Objectives

- Become familiar with the ways in which databases interact with information systems
- Develop skills in entity modeling
- Learn substantial parts of SQL
- Gain hands-on experience with MS Access
- Gain experience with MySQL

Grading

- Grade basis
  - Exams
  - Assignments (regular and extra credit)
  - Class questions
- Exam questions will be “easy” if you have carefully completed the assignments
- Grade distribution according to SBIAHU grading policies.

Grading

- Final grades are calculated based on a formula (no subjective grading)
- Formula weights all the components of the class
  - Exams/project
    - mid-term (20%), final (25%), project (25%), class-exercises/quizzes (15%), and homework (15%)
  - In-class exercises – points will be added if your team is the first to complete the exercise correctly
  - Assignments - No late submissions accepted.
  - Quiz questions will be taken primarily from textbook (lecture material) Questions
- Final grade is based on your total score in all grade components (the higher the score, the higher the grade)

- Grade Cutoffs

  A 95-100, A- 85-94, B+ 82-84, B 78-81, B- 75-77, C+ 72-74, C 68-71, C- 65-67, D 60-64, F 0-59

Exams

- Midterm Exam:

  On the weekend of week 8

- Final Exam:

  During university exam weeks (June 23 - July 2)

No make-up exams!

Assignments

- Assignments submissions
  - To receive credit there must be a reasonable attempt and at a minimum, SQL code must use the correct syntax
- The assignments must be done individually; group work will not be accepted.
- Assignments will be graded, and
  - You will get feedback
  - Submitting on-time counts to your grade
  - Components of the assignment problems will be used in exams

Assignments Deadlines

- Homework 1 – SQL Queries
  Deadline: Week 4
- Homework 2 – Multiple Tables
  Deadline: Week 7
- Homework 3 – Use of Wizard in MS-Access
  Deadline: Week 9
- Homework 4 – Relational Data Model
  Deadline: Week 11
- Project
  Deadlines:
  - Week 3: Submit the topic and list of team members.
  - Week 4: Write a statement of your database project. The summary should be between a half page and a page
  - Week 13: Submit the database and PowerPoint slides.
  - Week of 15-16: Present the project.
  - Week 17: Submit the finalized version of the project.

Note: Homework assignments must be completed individually. Group efforts are not allowed. However, projects can be done in groups of up to 3 students.

Academic Integrity

- University rules for academic integrity located at https://www.stonybrook.edu/commcms/academic_integrity/index.html
- All your work should be your own with the exception of:
  - Collaboration with approved teammates on project and designated assignments
  - Cited references to information sources in your project

Project

- There is one major project in the course
- The project covers most of the important concepts covered in the course
- Project:
  - Design a DB (on a topic of interest to you) and develop an implementation in MS Access
  - DB should have 5-10 entities
  - In-class presentations (RDM and final)
  - **Import a large amount of data**

Project Presentation

- Project presented by team (**in English**)
- 20 minutes
- Presentation plus operation of your DB
- Structured as an oral exam
- In-class or limited attendance presentation

Project Steps …

- More details during semester
- Form a group (**maximum of 3 in a group**)
- Write a statement of your database project
- Create an initial RDM for your project
- Revise your proposed project RDM according to the feedback you have received
- Create the corresponding tables (primarily from data import)
- Create simple forms to enter/modify data

… Project Steps

- Make up 5 "interesting" questions for your DB
- Create at least 2 hierarchical forms
- Create at least one report for your database
- Sign up for a time once you have your team
  - Better to sign up early
- Add 3-5 queries to your database to answer interesting questions

You will submit project components over the last month or so of the class – lateness counts

Lab Access and Software

- Software for the course should be available to you for your own computer
- If you need to use a University computer for assignments, studies, etc. you can use the Lab

Lab will run related software systems (MS Access, MS Visio)

DB Development Environment

- Assignments and class work
  - MS Access is used
  - MySQL will be introduced later (primarily using a non-programming interface)
- MS Access available on MSDNAA
- RDM Diagrams can be developed using any available tool, in MS Access, or MySQL Workbench

How to Get Help

- Don’t get stuck on a DB problem - ask for help
- Send an e-mail if you are having trouble
- Make an appointment or just stop by

We have TAs. Help will be available during class and for after-class sessions.

Please be sure to include “ISE305” in the subject line of any e-mail message to me

How to Learn the Material

- Don’t plan on memorizing material – plan on learning to think about the material – working problems is an important way to learn the material
- Plan on equal time spent reading text/notes and working hands-on exercises
  - Reading assignments will be provided
  - Read material in advance
- Work on the project in a small group (1-3 students)
- Attend class / review the on-line class notes

Lecture Slides

- Lecture slides will be available before each lecture
- Have a copy of the slides before class and use it to make notes
- Be sure to review the slides before each exam

Use of AI Tools

While AI-based resources can be helpful for learning, their use in completing programming assignments is strictly prohibited. Utilizing AI tools to generate or complete your code undermines the educational goals of this course and violates the university's academic integrity policy. All assignments must be your own work. Plagiarism or unauthorized assistance, including AI-generated code, will result in disciplinary action. Please remember that seeking help from teaching assistants, tutors, or study groups is encouraged and can be a valuable part of the learning process. However, all work submitted must be done independently to ensure you fully develop the skills needed for advanced courses and your future career. The skills you build now are crucial to your success, and adhering to these guidelines will help you achieve your academic and professional goals.

Lecture Schedule

Week Number	Monday Date	Lecture Dates/Topics	Note	Instructors
1	2025-02-17	02-18 Introduction, Background	Semester Start: First day of classes.	Cheng Zhang
2	2025-02-24	02-25 Single Entity - Basics		
3	2025-03-03	03-04 Single Entity - Other		
4	2025-03-10	03-11 Single Entity - GUI		
5	2025-03-17	03-18 One To Many		
6	2025-03-24	03-25 Multi-Table Queries		
7	2025-03-31	04-01 Workbench Modeling		
8	2025-04-07	04-08 Many To Many		
9	2025-04-14	04-15 One To One	Thu, Apr. 17th to Sat, Apr. 19th: University Sports Day, no classes in session.	
10	2025-04-21	04-22/04-24 Normalization & Examples		
11	2025-04-28	04-29 Data Modeling		
12	2025-05-05	05-06/05-08 Views & Forms	May 1-5, National Holiday, no classes in session.	
13	2025-05-12	05-13/05-15 Reports		
14	2025-05-19	05-20/05-22 MySQL		
15	2025-05-26	Dates TBD, Review Lectures, Check Final Group Project Presentations	May 31-June 2: National Holiday, no classes in session.	Ahmad Esmaili
16	2025-06-02			
17	2025-06-09	Dates TBD, - History and trends of database systems - Advanced SQL queries - Introduction of big data management systems		Fusheng Wang
18	2025-06-16		June 20: Classes End: Last day of classes	

Questions?